King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam I Term 261

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. For the triangle with vertices

$$A(-3,0,0), \qquad B(0,0,0), \qquad C(1,2,3),$$

which of the following is correct?

(a) Its type cannot be determined.

(b) It is an acute triangle.

(c) It is an equilateral triangle.

(d) It is an obtuse triangle.

(e) It is a right triangle.

2. Consider the parametric curve

$$x(t) = t^5 - 4t^3 + 3t + 4, \qquad y(t) = t^2 - 1.$$

At how many distinct points does the curve cross itself?

(a) 4

(b) 0

(c) 3

(d) 1

(e) 2

3. Which of the following polar equations does NOT represent a circle?

(a) $r = -1$

(b) $r = \sec\theta$

(c) $r = \sin\theta$

(d) $r = \cos\theta$

(e) $r = 2$

4. Which of the following statements is/are FALSE?

(I) If $\mathbf{u} = a\mathbf{i} + b\mathbf{j}$ is a unit vector, then $a^2 + b^2 = 1$.
(II) If $a + b = 0$, then $a\mathbf{i} + b\mathbf{j} = \mathbf{0}$.
(III) If $a = b$, then $\|a\mathbf{i} + b\mathbf{j}\| = \sqrt{2}\, a$.

(a) (I) and (II) only

(b) (II) and (III) only

(c) (II) only

(d) (I) only

(e) (III) only

5. The length of the parametric curve

$$x = \theta - \sin\theta, \qquad y = 1 - \cos\theta, \qquad 0 \le \theta \le 2\pi,$$

is:

(a) 8π

(b) $8\sqrt{2}$

(c) 4π

(d) 8

(e) 4

6. Given that $-\infty < t < \infty$, the set of parametric equations that represents exactly the same graph as

$$x = t, \qquad y = 2t + 1$$

is:

(a) $x = \cos t, \quad y = 2\cos t + 1$

(b) $x = t + 1, \quad y = 2t + 1$

(c) $x = -t^3, \quad y = -2t^3 + 1$

(d) $x = t^2, \quad y = 2t^2 + 1$

(e) $x = e^t, \quad y = 2e^t + 1$

7. The number of points of intersection of the polar curves

$$r = 3 - 2\sin\theta \qquad \text{and} \qquad r = 5\cos 2\theta$$

is:

(a) 7

(b) 6

(c) 4

(d) 8

(e) 5

8. The area of the surface generated by revolving the curve

$$x = 5\cos\theta, \qquad y = 5\sin\theta, \qquad 0 \le \theta \le \frac{\pi}{2}$$

about the y-axis is:

(a) 50π

(b) 10π

(c) 100π

(d) 25π

(e) 5π

9. The point that lies two-thirds of the way from $P(4, 3, 0)$ to $Q(1, -3, 3)$ is:

(a) $(3, 1, 2)$

(b) $(2, -1, 2)$

(c) $(3, -1, 2)$

(d) $(2, 0, 2)$

(e) $(2, -1, 1)$

10. Which of the following is NOT equal to $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$?

(a) $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{u})$

(b) $(\mathbf{v} \times \mathbf{w}) \cdot \mathbf{u}$

(c) $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$

(d) $\mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})$

(e) $\mathbf{u} \cdot (\mathbf{w} \times \mathbf{v})$

11. Find the area of the region common to the two regions bounded by

$$r = 2\cos\theta \qquad \text{and} \qquad r = 2\sin\theta.$$

(a) $\dfrac{\pi}{2} + 1$

(b) $\dfrac{\pi}{4}$

(c) $\pi - 2$

(d) π

(e) $\dfrac{\pi}{2} - 1$

12. The volume of the parallelepiped that has

$$\mathbf{u} = 3\mathbf{i} - 5\mathbf{j} + \mathbf{k}, \qquad \mathbf{v} = 2\mathbf{j} - 2\mathbf{k}, \qquad \mathbf{w} = 3\mathbf{i} + \mathbf{j} + \mathbf{k}$$

as adjacent edges is:

(a) 42

(b) 24

(c) 18

(d) 30

(e) 36

13. If $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w}$ and $\mathbf{u} \neq \mathbf{0}$, which of the following must be TRUE?

(a) $\|\mathbf{v}\| = \|\mathbf{w}\|$

(b) $\mathbf{v} = \mathbf{w}$

(c) $\mathbf{v}$ is perpendicular to $\mathbf{w}$

(d) $\mathbf{v} + \mathbf{w}$ is perpendicular to $\mathbf{u}$

(e) $\mathbf{v} - \mathbf{w}$ is perpendicular to $\mathbf{u}$

14. The rectangular equation of the curve

$$x = \cos\theta, \qquad y = 2\sin 2\theta, \qquad 0 \leq \theta \leq \frac{\pi}{2}$$

is:

(a) $y = 2x\sqrt{1 - x^2}$

(b) $y = -2x\sqrt{1 - x^2}$

(c) $y = -2x\sqrt{1 - x}$

(d) $y = 4x\sqrt{1 - x^2}$

(e) $y = 2x\sqrt{1 - x}$

15. The equation of the tangent line to the curve

$$x = t^4 + 2, \qquad y = t^3 + t$$

at the point $(3, -2)$ is:

(a) $y = x - 5$

(b) $y = -x + 5$

(c) $y = x + 1$

(d) $y = -x + 1$

(e) $y = -2x + 4$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$\overrightarrow{BA} \cdot \overrightarrow{BC} = -3 < 0$, so the angle at B is obtuse.
2	**(e)**	$x(t) = x(-t)$, $t \neq 0$: $t = \pm 1, \pm\sqrt{3}$, so 2 crossing points.
3	**(b)**	$r = \sec\theta$ means $x = 1$, a line; the other four convert to circles.
4	**(b)**	(I) true; (II) false ($\mathbf{i} - \mathbf{j}$); (III) false (norm is $\sqrt{2}\lvert a\rvert$).
5	**(d)**	$ds = 2\sin\frac{\theta}{2}\,d\theta$; $L = \int_0^{2\pi} 2\sin\frac{\theta}{2}\,d\theta = 8$.
6	**(c)**	Need $y = 2x + 1$ and all real x: only $x = -t^3$, $y = -2t^3 + 1$ works.
7	**(a)**	$5s^2 - s - 1 = 0$ gives 4 points; $5s^2 + s - 4 = 0$ gives 3 more: total 7.
8	**(a)**	$S = 2\pi \int_0^{\pi/2} 5\cos\theta \cdot 5\,d\theta = 50\pi$.
9	**(b)**	$P + \frac{2}{3}(Q - P) = (4, 3, 0) + (-2, -4, 2) = (2, -1, 2)$.
10	**(e)**	Swapping $\mathbf{v}, \mathbf{w}$ in the cross product flips the sign; the others are cyclic.
11	**(e)**	Meet at $\theta = \frac{\pi}{4}$; $A = 2\int_0^{\pi/4}(1 - \cos 2\theta)\,d\theta = \frac{\pi}{2} - 1$.
12	**(e)**	$V = \lvert\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})\rvert = \lvert 12 + 30 - 6\rvert = 36$.
13	**(e)**	$\mathbf{u} \cdot (\mathbf{v} - \mathbf{w}) = 0$, so $\mathbf{v} - \mathbf{w} \perp \mathbf{u}$.
14	**(d)**	$y = 4\sin\theta\cos\theta$ with $\sin\theta = \sqrt{1 - x^2}$: $y = 4x\sqrt{1 - x^2}$.
15	**(d)**	$t = -1$, slope $\frac{3t^2+1}{4t^3} = -1$: $y + 2 = -(x - 3)$, so $y = -x + 1$.